

MTH304 - Endsem

Each problem is worth 5 marks. Attempt any 10 of the 14 provided.

1. Recall the definition of the commutator subgroup $G^{(1)}$ of G . Show that $G^{(1)}$ is normal in G . Show that $G/G^{(1)}$ is Abelian.
2. Show that S_3 is a solvable but not nilpotent group.
3. List all Abelian Groups of order 36. (Context: We are allowed to use the Structure Theorem without proof)
4. Let H, N be subgroups of a group G , and assume N is normal in G . Let $p : G \rightarrow G/N$ be the quotient homomorphism.
 - (a) Determine the kernels of the restriction of p to H and HN
 - (b) Apply the first isomorphism theorem to these restrictions to prove

$$H/(H \cap N) = (HN)/N$$

5. Consider the subgroup $B \subset GL_2(\mathbb{C})$ by

$$B := \left\{ \begin{pmatrix} a & c \\ 0 & b \end{pmatrix} \mid a, b, c \in \mathbb{C}, ab \neq 0 \right\}$$

We set

$$U := \left\{ \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix} \mid c \in \mathbb{C} \right\} \quad D := \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{C}, ab \neq 0 \right\}$$

- (a) Show that U is a normal subgroup of B
 - (b) Describe the conjugation action of D on U
 - (c) Show that $B = U \rtimes D$
6. Extend the set map

$$\begin{aligned} \Phi : \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} &\rightarrow (\mathbb{Z}/16\mathbb{Z})^* \\ (1, 0) &\mapsto 3 \\ (0, 1) &\mapsto -1 = 15 \end{aligned}$$

by defining the image of $(x, y) \in \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ in a way that one gets a group homomorphism. Show that Φ is an isomorphism.

7. Consider $N \subset GL_n(\mathbb{C})$ consisting of matrices with exactly one nonzero entry in each row and column. Define a group homomorphism from N to S_n and describe the kernel T . Describe N in terms of T and S_n .
8. Show that the alternating group A_n is generated by 3-cycles. Deduce that the commutator subgroup of S_n is A_n .

9. Show that the centre of S_n is trivial for $n \geq 3$. Deduce that S_n is not the direct product $A_n \times \mathbb{Z}/2\mathbb{Z}$.
10. Show that a group of order 56 is not simple.
11. Show that a group of order 33 is Abelian.
12. In the 3-space $\mathbb{R}^3$, write the matrix that rotates by an angle $\theta \in [0, 2\pi)$ in the anti-clockwise direction keeping the Z -axis fixed. Do the same keeping the Y -axis fixed. Write the matrix corresponding to the reflection in the YZ -plane.
13. Consider the conjugation action of S_5 on itself. Describe the centraliser subgroup of $\begin{pmatrix} 1 & 2 \end{pmatrix}$ by first computing its order. (Context: We need to do this for each S_n and thus, deduce the structure of the centraliser as a function of n)
14. Let G be a group and we denote the underlying set as S . Consider the left-translation action of a group G on itself i.e.

$$\begin{aligned} G \times S &\rightarrow S \\ (g, s) &\mapsto g * s := gs \end{aligned}$$

For $s \in S$, describe the group $\text{Stab}(s)$ and the set $\text{Orb}(s)$. Verify the Orbit-Stabiliser Theorem. Consider the conjugation action of a group G on itself, i.e.

$$\begin{aligned} G \times S &\rightarrow S \\ (g, s) &\mapsto g * s := gsg^{-1} \end{aligned}$$

Assume that G is abelian. For $s \in S$, describe the group $\text{Stab}(s)$ and the set $\text{Orb}(s)$. Verify the Orbit-Stabiliser Theorem.